

1st Part Problem Set for Introduction to Jordan algebras

1. Let A be an associative algebra. Consider $B = A \oplus A^{op}$, where $A^{op} = (A, a \cdot^{op} b := ba)$, an algebra with an involution $(a, b)^* = (b, a)$. Show that the Jordan algebras are isomorphic

$$H(B, *) \simeq A^+.$$

2. Let A be an associative algebra with involution $*$. If $(A, *)$ is $*$ -simple algebra (i.e. $A \neq 0$ and there are no non-trivial ideal I in A , with $I^* = I$), then $H(A, *)$ is simple.
3. Let $J(V, f)$ be a Jordan algebra of Clifford type. Show that
- a) if $\dim V > 1$ and f is non-degenerated then $J(V, f)$ is simple algebra;
 - b) $J(V, f)$ is special (*Hint: use embedding into the Clifford algebra of f*).
4. An algebra A is called *power associative* if $\text{alg}\langle a \rangle$ is associative algebra for any $a \in A$. Show that any Jordan algebra is power associative algebra.
5. Give an example of an associative non-commutative algebra A such that A^+ is associative as well.
6. Let A be an associative algebra such that A^+ is associative as well. Prove that A is then PI algebra (algebra whose elements satisfy non-trivial polynomial identity). Determine which identity is it.
7. Let A be commutative associative algebra with a derivation D satisfying $D^3 = 0$. Define a new multiplication in A

$$a \star b = D(a)D(b).$$

Show that $(A, \star)$ is a Jordan algebra.

8. For any Jordan algebra J show that there exists a Jordan algebra $B \supseteq J$ such that the universal multiplicative envelope $\mathcal{U}(J) \simeq \text{Mult}_B(J)$
9. Let J be a Jordan algebra. Show that if $\text{Mult}(J)^n = 0$ then $J^{2n-1} = 0$.
10. Let A be a finite-dimensional algebra. Show that A contains solvable ideal $\text{Solv}(A)$, which in turn contains all solvable ideals in A , moreover $\text{Solv}(A/\text{Solv}A) = 0$.
11. Give an example of Jordan algebra J and its ideal I such that I^2 is not an ideal of J .
12. Prove that in any commutative or skew-commutative algebra A the following identities are satisfied for any $a, b, c \in A$

$$(a, b, a) = 0 \quad (a, b, c) = -(c, b, a).$$

13. Show that for any finite-dimensional Jordan algebra J its nil-radical is hereditary, namely for any ideal I of J we have $Nil I = I \cap Nil J$.
14. Verify that in Jordan algebra J we have $U_{a,b} = R_a R_b + R_b R_a - R_{ab}$, $a, b \in J$.
15. Show that for special Jordan algebra $(A^+, a \cdot b = \frac{1}{2}(ab + ba))$, $\{a, b, c\} = \frac{1}{2}(abc + cba)$.
16. *The full Peirce Decomposition* Let J be Jordan algebra with unity element 1. Let $e_1, \dots, e_n$ be non-zero, pairwise orthogonal idempotents satisfying $e_1 + \dots + e_n = 1$. Prove that

$$J = \bigoplus_{i=1}^n J_{ii} \oplus \left(\bigoplus_{1 \leq i < j \leq n} J_{ij} \right),$$

where $J_{ii} = J_1(e_i)$ and $J_{ij} = J_{\frac{1}{2}}(e_i) \cap J_{\frac{1}{2}}(e_j)$ for $1 \leq i < j \leq n$.

17. Show that Peirce decomposition components satisfy the following relations:

$$\begin{aligned} J_{ii}^2 &\subseteq J_{ii}, & J_{ij}^2 &\subseteq J_{ii} + J_{jj}, & J_{ii}J_{ij} &\subseteq J_{ij}, & J_{ii}J_{jj} &= 0 \quad i \neq j, \\ J_{ij}J_{jk} &\subseteq J_{ik} \quad (i, j, k \text{ distinct}), & J_{ij}J_{kl} &= J_{ii}J_{kl} \quad (i, j, k, l \text{ distinct}). \end{aligned}$$

18. Suppose that J has the full Peirce decomposition. Show that $J_{ij}^2 e_i$ is an ideal in J_{ii} .
19. Consider polynomial algebra $F[x_1, \dots, x_n]$ with a derivations $D(f) = \sum_{i=1}^n \frac{\partial f}{\partial x_i}$ and its subspace $J = \{f \mid f \text{ homogeneous, } \deg f = 2\}$. Prove that J is Jordan algebra with respect to the product $f \star g = D(f)D(g)$. Prove that J is finite-dimensional and simple. Which simple finite-dimensional Jordan algebra from classification theorem $(J, \star)$ corresponds to?
20. Prove that simple Jordan algebra $J(V, f)$ over an algebraically closed field is an algebra of capacity 2.
21. Prove that $J(V, f)$ contains non-trivial idempotent if and only if there exists $x \in V$ with $f(x, x) = 1$.
22. Prove that every commutative nilpotent algebra of dimension three is Jordan.
23. An algebra A is called locally nilpotent if every finitely generated subalgebra of A is nilpotent. Prove that simple algebra is not locally nilpotent.
24. Prove that for octonion algebra $\mathbb{O}$, $\mathbb{O}^+$ is isomorphic to a Jordan algebra of a symmetric nondegenerate bilinear form.
25. Let J be finite dimensional Jordan algebra over a field F and $R_a : x \rightarrow xa$ be the operator of right multiplication in J . Show that
- (i) if a is nilpotent $tr(R_a) = 0$,
 - (ii) if $char F = 0$ and e is an idempotent in J then $tr R_e \neq 0$.
26. Describe all Jordan algebras of dimension two up to an isomorphism.